

Experiment 2: Random Stream Data Processing Pipeline

Aim

To generate a random data stream and visualize it using line and scatter plots.

Theory

Random data is generated to simulate a stream and visualized using Matplotlib.

Requirements

- Python 3.x
- Jupyter Notebook
- Matplotlib

Step 1: Import Required Libraries

```
import random
import matplotlib.pyplot as plt
```

Interpretation: Imports required libraries.

Step 2: Generate Random Stream Data

```
stream_data=[]
for i in range(100):
    value=random.randint(1,100)
    stream_data.append(value)
```

Interpretation: Generates 100 random values.

Step 3: Line Plot Visualization

```
plt.plot(stream_data)
plt.title('Random Stream Data')
plt.xlabel('Data Point Number')
plt.ylabel('Random Value')
plt.grid(True)
plt.show()
```

Interpretation: Displays the line plot.

Step 4: Scatter Plot Visualization

```
x=[]
for i in range(100):
    x.append(i+1)
plt.scatter(x,stream_data)
plt.title('2D View of Stream Data')
plt.xlabel('Data Point Number')
plt.ylabel('Random Value')
plt.grid(True)
plt.show()
```

Interpretation: Displays the scatter plot.

Complete Program

```
import random
import matplotlib.pyplot as plt

stream_data = []
```

```
for i in range(100):
    value = random.randint(1, 100)
    stream_data.append(value)

plt.plot(stream_data)
plt.title("Random Stream Data")
plt.xlabel("Data Point Number")
plt.ylabel("Random Value")
plt.grid(True)
plt.show()

x = []

for i in range(100):
    x.append(i + 1)

plt.scatter(x, stream_data)
plt.title("2D View of Stream Data")
plt.xlabel("Data Point Number")
plt.ylabel("Random Value")
plt.grid(True)
plt.show()
```

Expected Output

A line plot and a scatter plot showing 100 random values.

Result

The random stream data was successfully generated and visualized using line and scatter plots.